Mathematical Supplement: Duality Theory with Cobb-Douglas Example

Lecture 2 - Demand Analysis and Consumer Welfare

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Introduction

This document provides complete mathematical derivations for all duality results covered in Lecture 2. We use the **Cobb-Douglas utility function** as a running example throughout:

$$U(x, y) = x^{\alpha} y^{\beta}$$

where $\alpha, \beta > 0$. We assume the consumer faces prices (p_x, p_y) and has income I.

1. The Primal Problem: Utility Maximization

1.1 Problem Statement

The consumer's problem is:

$$\max_{x,y} \quad x^\alpha y^\beta$$

subject to:

$$p_x x + p_y y = I$$
$$x \geq 0, \quad y \geq 0$$

1.2 Lagrangian Method

Set up the Lagrangian:

$$\mathcal{L}(x, y, \lambda) = x^\alpha y^\beta + \lambda(I - p_x x - p_y y)$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = \alpha x^{\alpha-1} y^\beta - \lambda p_x = 0 \tag{1}$$
$$\frac{\partial \mathcal{L}}{\partial y} = \beta x^\alpha y^{\beta-1} - \lambda p_y = 0 \tag{2}$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - p_x x - p_y y = 0 \tag{3}$$

1.3 Solving for Demand Functions

From equations (1) and (2):

$$\frac{\alpha x^{\alpha-1} y^\beta}{\beta x^\alpha y^{\beta-1}} = \frac{p_x}{p_y}$$

Simplifying:

$$\frac{\alpha y}{\beta x} = \frac{p_x}{p_y}$$

Solving for y:

$$y = \frac{\beta p_x}{\alpha p_y} x \tag{4}$$

Substitute equation (4) into the budget constraint (3):

$$p_x x + p_y \cdot \frac{\beta p_x}{\alpha p_y} x = I$$

$$p_x x \left(1 + \frac{\beta}{\alpha}\right) = I$$

$$p_x x \cdot \frac{\alpha + \beta}{\alpha} = I$$

Solving for x:

$$\boxed{x^*(p_x, p_y, I) = \frac{\alpha I}{(\alpha+\beta)p_x}}$$

Substituting back into equation (4):

$$y^* = \frac{\beta p_x}{\alpha p_y} \cdot \frac{\alpha I}{(\alpha+\beta)p_x} = \frac{\beta I}{(\alpha+\beta)p_y}$$

$$\boxed{y^*(p_x, p_y, I) = \frac{\beta I}{(\alpha+\beta)p_y}}$$

1.4 Properties of Marshallian Demand

Expenditure shares:

$$\frac{p_x x^*}{I} = \frac{p_x \cdot \alpha I/[(\alpha+\beta)p_x]}{I} = \frac{\alpha}{\alpha+\beta}$$

$$\frac{p_y y^*}{I} = \frac{\beta}{\alpha+\beta}$$

Expenditure shares are constant, independent of prices and income.

Homogeneity of degree zero:

$$x^*(tp_x, tp_y, tI) = \frac{\alpha \cdot tI}{(\alpha+\beta)tp_x} = \frac{\alpha I}{(\alpha+\beta)p_x} = x^*(p_x, p_y, I)$$

2. Indirect Utility Function

2.1 Definition

The **indirect utility function** gives maximum utility as a function of prices and income:

$$V(p_x, p_y, I) = U(x^*(p_x, p_y, I), y^*(p_x, p_y, I))$$

2.2 Derivation for Cobb-Douglas

Substitute the demand functions into the utility function:

$$V(p_x, p_y, I) = \left(\frac{\alpha I}{(\alpha+\beta)p_x}\right)^{\alpha} \left(\frac{\beta I}{(\alpha+\beta)p_y}\right)^{\beta}$$

Simplifying:

$$V(p_x, p_y, I) = \frac{\alpha^\alpha \beta^\beta}{(\alpha+\beta)^{\alpha+\beta}} \cdot \frac{I^{\alpha+\beta}}{p_x^\alpha p_y^\beta}$$

Let $K = \frac{\alpha^\alpha \beta^\beta}{(\alpha+\beta)^{\alpha+\beta}}$:

$$\boxed{V(p_x, p_y, I) = K \cdot \frac{I^{\alpha+\beta}}{p_x^\alpha p_y^\beta}}$$

2.3 Properties

Increasing in income:

$$\frac{\partial V}{\partial I} = K \cdot (\alpha+\beta)\frac{I^{\alpha+\beta-1}}{p_x^\alpha p_y^\beta} > 0$$

Decreasing in prices:

$$\frac{\partial V}{\partial p_x} = K \cdot I^{\alpha+\beta} \cdot (-\alpha)\frac{p_x^{-\alpha-1}}{p_y^\beta} < 0$$

Homogeneous of degree zero:

$$V(tp_x, tp_y, tI) = K \cdot \frac{(tI)^{\alpha+\beta}}{(tp_x)^\alpha (tp_y)^\beta} = K \cdot \frac{t^{\alpha+\beta} I^{\alpha+\beta}}{t^{\alpha+\beta} p_x^\alpha p_y^\beta} = V(p_x, p_y, I)$$

3. The Dual Problem: Expenditure Minimization

3.1 Problem Statement

The dual problem is:

$$\min_{x,y} \quad p_x x + p_y y$$

subject to:

$$x^\alpha y^\beta = \bar{U}$$

$$x \geq 0, \quad y \geq 0$$

where $\bar{U}$ is a target utility level.

3.2 Lagrangian Method

Set up the Lagrangian:

$$\mathcal{L}(x, y, \mu) = p_x x + p_y y + \mu(\bar{U} - x^\alpha y^\beta)$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = p_x - \mu \alpha x^{\alpha-1} y^\beta = 0 \tag{5}$$

$$\frac{\partial \mathcal{L}}{\partial y} = p_y - \mu \beta x^\alpha y^{\beta-1} = 0 \tag{6}$$

$$\frac{\partial \mathcal{L}}{\partial \mu} = \bar{U} - x^\alpha y^\beta = 0 \tag{7}$$

3.3 Solving for Hicksian Demand

From equations (5) and (6):

$$\frac{p_x}{p_y} = \frac{\mu \alpha x^{\alpha-1} y^\beta}{\mu \beta x^\alpha y^{\beta-1}} = \frac{\alpha y}{\beta x}$$

Solving for y:

$$y = \frac{\beta p_x}{\alpha p_y} x \tag{8}$$

Substitute equation (8) into the utility constraint (7):

$$x^\alpha \left(\frac{\beta p_x}{\alpha p_y} x \right)^\beta = \bar{U}$$

$$x^{\alpha+\beta} \left(\frac{\beta p_x}{\alpha p_y} \right)^\beta = \bar{U}$$

$$x^{\alpha+\beta} = \bar{U} \left(\frac{\alpha p_y}{\beta p_x} \right)^\beta$$

$$x = \bar{U}^{1/(\alpha+\beta)} \left(\frac{\alpha p_y}{\beta p_x} \right)^{\beta/(\alpha+\beta)}$$

Rearranging:

$$\boxed{x^h(p_x, p_y, \bar{U}) = \left(\frac{\alpha}{\beta} \right)^{\beta/(\alpha+\beta)} \left(\frac{p_y}{p_x} \right)^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}}$$

Similarly, substituting back:

$$\boxed{y^h(p_x, p_y, \bar{U}) = \left(\frac{\beta}{\alpha} \right)^{\alpha/(\alpha+\beta)} \left(\frac{p_x}{p_y} \right)^{\alpha/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}}$$

3.4 Expenditure Function

The expenditure function is:

$$E(p_x, p_y, \bar{U}) = p_x x^h + p_y y^h$$

Substituting the Hicksian demands:

$$E = p_x \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} \left(\frac{p_y}{p_x}\right)^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)} + p_y \left(\frac{\beta}{\alpha}\right)^{\alpha/(\alpha+\beta)} \left(\frac{p_x}{p_y}\right)^{\alpha/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$$

Simplifying the first term:

$$p_x \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} \left(\frac{p_y}{p_x}\right)^{\beta/(\alpha+\beta)} = \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)}$$

Similarly for the second term:

$$p_y \left(\frac{\beta}{\alpha}\right)^{\alpha/(\alpha+\beta)} \left(\frac{p_x}{p_y}\right)^{\alpha/(\alpha+\beta)} = \left(\frac{\beta}{\alpha}\right)^{\alpha/(\alpha+\beta)} p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)}$$

Therefore:

$$E = p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)} \left[\left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} + \left(\frac{\beta}{\alpha}\right)^{\alpha/(\alpha+\beta)} \right]$$

Let:

$$C = \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} + \left(\frac{\beta}{\alpha}\right)^{\alpha/(\alpha+\beta)}$$

Then:

$$\boxed{E(p_x, p_y, \bar{U}) = C \cdot p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}}$$

3.5 Properties of Expenditure Function

Increasing in $\bar{U}$:

$$\frac{\partial E}{\partial \bar{U}} = C \cdot p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \cdot \frac{1}{\alpha+\beta} \bar{U}^{1/(\alpha+\beta)-1} > 0$$

Increasing in prices:

$$\frac{\partial E}{\partial p_x} = C \cdot \frac{\alpha}{\alpha+\beta} p_x^{\alpha/(\alpha+\beta)-1} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)} > 0$$

Homogeneous of degree 1 in prices:

$$E(tp_x, tp_y, \bar{U}) = C \cdot (tp_x)^{\alpha/(\alpha+\beta)} (tp_y)^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$$

$$= C \cdot t^{(\alpha+\beta)/(\alpha+\beta)} p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)} = t \cdot E(p_x, p_y, \bar{U})$$

4. Duality Relationships

4.1 Relating V and E

The indirect utility function and expenditure function are inverses:

Property 1: $V(p_x, p_y, E(p_x, p_y, \bar{U})) = \bar{U}$

Verification:

$$V(p_x, p_y, E) = K \cdot \frac{E^{\alpha+\beta}}{p_x^\alpha p_y^\beta}$$

Substituting $E = C \cdot p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$:

$$V = K \cdot \frac{\left[C \cdot p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}\right]^{\alpha+\beta}}{p_x^\alpha p_y^\beta}$$

$$= K \cdot \frac{C^{\alpha+\beta} \cdot p_x^\alpha p_y^\beta \bar{U}}{p_x^\alpha p_y^\beta} = KC^{\alpha+\beta}\bar{U}$$

For this to equal $\bar{U}$, we need $KC^{\alpha+\beta} = 1$, which can be verified algebraically.

Property 2: $E(p_x, p_y, V(p_x, p_y, I)) = I$

Verification:

$$E(p_x, p_y, V) = C \cdot p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} V^{1/(\alpha+\beta)}$$

Substituting $V = K \cdot \frac{I^{\alpha+\beta}}{p_x^\alpha p_y^\beta}$:

$$E = C \cdot p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \left[K \cdot \frac{I^{\alpha+\beta}}{p_x^\alpha p_y^\beta}\right]^{1/(\alpha+\beta)}$$

$$= C \cdot p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \cdot K^{1/(\alpha+\beta)} \cdot \frac{I}{p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)}}$$

$$= CK^{1/(\alpha+\beta)} I$$

For this to equal I, we need $CK^{1/(\alpha+\beta)} = 1$.

4.2 Relating Marshallian and Hicksian Demands

At the optimum:

$$x^*(p_x, p_y, I) = x^h(p_x, p_y, V(p_x, p_y, I))$$

Verification: We need to show:

$$\frac{\alpha I}{(\alpha+\beta)p_x} = \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} \left(\frac{p_y}{p_x}\right)^{\beta/(\alpha+\beta)} \left[K \frac{I^{\alpha+\beta}}{p_x^\alpha p_y^\beta}\right]^{1/(\alpha+\beta)}$$

The right-hand side simplifies to:

$$\left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)}\left(\frac{p_y}{p_x}\right)^{\beta/(\alpha+\beta)}K^{1/(\alpha+\beta)}\frac{I}{p_x^{\alpha/(\alpha+\beta)}p_y^{\beta/(\alpha+\beta)}}$$

$$=\left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)}K^{1/(\alpha+\beta)}\frac{I}{p_x^{\alpha/(\alpha+\beta)}p_y^{\beta/(\alpha+\beta)}}\cdot\frac{p_y^{\beta/(\alpha+\beta)}}{p_x^{\beta/(\alpha+\beta)}}$$

$$=\left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)}K^{1/(\alpha+\beta)}\frac{I}{p_x}$$

For equality, we need:

$$\left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)}K^{1/(\alpha+\beta)}=\frac{\alpha}{\alpha+\beta}$$

This holds by construction of K.

5. Shephard's Lemma

5.1 Statement

Shephard's Lemma: The partial derivative of the expenditure function with respect to a price equals the Hicksian demand:

$$\frac{\partial E(p_x, p_y, \bar{U})}{\partial p_x} = x^h(p_x, p_y, \bar{U})$$

5.2 Verification for Cobb-Douglas

We have:

$$E(p_x, p_y, \bar{U}) = C \cdot p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$$

Taking the derivative:

$$\frac{\partial E}{\partial p_x} = C \cdot \frac{\alpha}{\alpha+\beta} p_x^{\alpha/(\alpha+\beta)-1} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$$

$$= \frac{\alpha}{\alpha+\beta} \cdot \frac{C p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}}{p_x}$$

$$= \frac{\alpha}{\alpha+\beta} \cdot \frac{E(p_x, p_y, \bar{U})}{p_x}$$

Now compare with Hicksian demand. We need to verify this equals:

$$x^h = \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} \left(\frac{p_y}{p_x}\right)^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$$

From $E = p_x x^h + p_y y^h$ and the fact that expenditure shares are $\frac{\alpha}{\alpha+\beta}$ and $\frac{\beta}{\alpha+\beta}$:

$$p_x x^h = \frac{\alpha}{\alpha+\beta} E$$

Therefore:

$$x^h = \frac{\alpha}{\alpha+\beta} \cdot \frac{E}{p_x} = \frac{\partial E}{\partial p_x}$$

6. Roy's Identity

6.1 Statement

Roy's Identity: Marshallian demand can be recovered from the indirect utility function:

$$x(p_x, p_y, I) = -\frac{\partial V/\partial p_x}{\partial V/\partial I}$$

6.2 Derivation

Start with the identity:

$$V(p_x, p_y, E(p_x, p_y, \bar{U})) = \bar{U}$$

Differentiate both sides with respect to p_x:

$$\frac{\partial V}{\partial p_x} + \frac{\partial V}{\partial E} \cdot \frac{\partial E}{\partial p_x} = 0$$

By Shephard's Lemma, $\frac{\partial E}{\partial p_x} = x^h$. Also, note that $\frac{\partial V}{\partial E} = \frac{\partial V}{\partial I}$ (by chain rule). Therefore:

$$\frac{\partial V}{\partial p_x} + \frac{\partial V}{\partial I} \cdot x^h = 0$$

Solving for x^h:

$$x^h = -\frac{\partial V/\partial p_x}{\partial V/\partial I}$$

Since $x = x^h$ at the optimum when $E = I$:

$$x = -\frac{\partial V/\partial p_x}{\partial V/\partial I}$$

6.3 Verification for Cobb-Douglas

We have:

$$V(p_x, p_y, I) = K \cdot \frac{I^{\alpha+\beta}}{p_x^\alpha p_y^\beta}$$

Computing derivatives:

$$\frac{\partial V}{\partial p_x} = K \cdot I^{\alpha+\beta} \cdot (-\alpha) p_x^{-\alpha-1} p_y^{-\beta} = -\frac{\alpha V}{p_x}$$

$$\frac{\partial V}{\partial I} = K \cdot (\alpha+\beta) I^{\alpha+\beta-1} p_x^{-\alpha} p_y^{-\beta} = \frac{(\alpha+\beta)V}{I}$$

Applying Roy's Identity:

$$x = -\frac{-\alpha V/p_x}{(\alpha+\beta)V/I} = \frac{\alpha I}{(\alpha+\beta)p_x}$$

This matches our Marshallian demand!

7. The Slutsky Equation

7.1 Statement

The Slutsky equation decomposes the total effect of a price change into substitution and income effects:

$$\frac{\partial x}{\partial p_x} = \underbrace{\frac{\partial x^h}{\partial p_x}}_{\text{Substitution Effect}} - \underbrace{x \cdot \frac{\partial x}{\partial I}}_{\text{Income Effect}}$$

7.2 Derivation

Start with the identity:

$$x(p_x, p_y, I) = x^h(p_x, p_y, V(p_x, p_y, I))$$

Differentiate both sides with respect to p_x:

$$\frac{\partial x}{\partial p_x} = \frac{\partial x^h}{\partial p_x} + \frac{\partial x^h}{\partial \bar{U}} \cdot \frac{\partial V}{\partial p_x}$$

From Roy's Identity: $\frac{\partial V}{\partial p_x} = -x\frac{\partial V}{\partial I}$

Also, by the envelope theorem and duality:

$$\frac{\partial x^h}{\partial \bar{U}} = \frac{\partial x}{\partial I} \cdot \left(\frac{\partial V}{\partial I}\right)^{-1}$$

Substituting:

$$\frac{\partial x}{\partial p_x} = \frac{\partial x^h}{\partial p_x} + \frac{\partial x}{\partial I} \cdot \frac{1}{\partial V/\partial I} \cdot \left(-x\frac{\partial V}{\partial I}\right)$$

$$= \frac{\partial x^h}{\partial p_x} - x\frac{\partial x}{\partial I}$$

7.3 Verification for Cobb-Douglas

Marshallian demand:

$$x(p_x, p_y, I) = \frac{\alpha I}{(\alpha+\beta)p_x}$$

$$\frac{\partial x}{\partial p_x} = -\frac{\alpha I}{(\alpha+\beta)p_x^2}$$

$$\frac{\partial x}{\partial I} = \frac{\alpha}{(\alpha+\beta)p_x}$$

Hicksian demand:

$$x^h(p_x, p_y, \bar{U}) = \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} \left(\frac{p_y}{p_x}\right)^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$$

At the optimum, $\bar{U} = V(p_x, p_y, I) = K\frac{I^{\alpha+\beta}}{p_x^\alpha p_y^\beta}$, so:

$$x^h = \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} \left(\frac{p_y}{p_x}\right)^{\beta/(\alpha+\beta)} \left[K\frac{I^{\alpha+\beta}}{p_x^\alpha p_y^\beta}\right]^{1/(\alpha+\beta)}$$

$$= \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} K^{1/(\alpha+\beta)}\frac{I}{p_x}$$

Taking the derivative (holding $\bar{U}$ constant, so treat I as adjusting to keep utility constant):

$$\left.\frac{\partial x^h}{\partial p_x}\right|_{\bar{U}} = -\left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} K^{1/(\alpha+\beta)}\frac{I}{p_x^2}$$

But at the optimum $x = \frac{\alpha I}{(\alpha+\beta)p_x}$, so:

$$\left.\frac{\partial x^h}{\partial p_x}\right|_{\bar{U}} = -\frac{\beta}{\alpha+\beta} \cdot \frac{x}{p_x}$$

Verifying the Slutsky equation:

$$\frac{\partial x}{\partial p_x} = -\frac{\alpha I}{(\alpha+\beta)p_x^2}$$

$$\frac{\partial x^h}{\partial p_x} - x\frac{\partial x}{\partial I} = -\frac{\beta}{\alpha+\beta} \cdot \frac{x}{p_x} - \frac{\alpha I}{(\alpha+\beta)p_x} \cdot \frac{\alpha}{(\alpha+\beta)p_x}$$

$$= -\frac{\beta x}{(\alpha+\beta)p_x} - \frac{\alpha^2 I}{(\alpha+\beta)^2 p_x^2}$$

Substituting $x = \frac{\alpha I}{(\alpha+\beta)p_x}$:

$$= -\frac{\beta\alpha I}{(\alpha+\beta)^2 p_x^2} - \frac{\alpha^2 I}{(\alpha+\beta)^2 p_x^2} = -\frac{\alpha(\alpha+\beta)I}{(\alpha+\beta)^2 p_x^2} = -\frac{\alpha I}{(\alpha+\beta)p_x^2}$$

This matches $\frac{\partial x}{\partial p_x}$!

8. Summary of All Results

For the Cobb-Douglas utility function $U(x, y) = x^\alpha y^\beta$:

8.1 Marshallian Demand Functions

$$x^*(p_x, p_y, I) = \frac{\alpha I}{(\alpha+\beta)p_x}, \quad y^*(p_x, p_y, I) = \frac{\beta I}{(\alpha+\beta)p_y}$$

8.2 Indirect Utility Function

$$V(p_x, p_y, I) = K \cdot \frac{I^{\alpha+\beta}}{p_x^\alpha p_y^\beta}$$

where $K = \frac{\alpha^\alpha \beta^\beta}{(\alpha+\beta)^{\alpha+\beta}}$

8.3 Hicksian Demand Functions

$$x^h(p_x, p_y, \bar{U}) = \left(\frac{\alpha}{\beta}\right)^{\beta/(\alpha+\beta)} \left(\frac{p_y}{p_x}\right)^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$$

$$y^h(p_x, p_y, \bar{U}) = \left(\frac{\beta}{\alpha}\right)^{\alpha/(\alpha+\beta)} \left(\frac{p_x}{p_y}\right)^{\alpha/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$$

8.4 Expenditure Function

$$E(p_x, p_y, \bar{U}) = C \cdot p_x^{\alpha/(\alpha+\beta)} p_y^{\beta/(\alpha+\beta)} \bar{U}^{1/(\alpha+\beta)}$$

8.5 Key Identities

Shephard's Lemma:

$$\frac{\partial E}{\partial p_x} = x^h, \quad \frac{\partial E}{\partial p_y} = y^h$$

Roy's Identity:

$$x = -\frac{\partial V/\partial p_x}{\partial V/\partial I}, \quad y = -\frac{\partial V/\partial p_y}{\partial V/\partial I}$$

Slutsky Equation:

$$\frac{\partial x}{\partial p_x} = \frac{\partial x^h}{\partial p_x} - x\frac{\partial x}{\partial I}$$

Duality:

$$V(p_x, p_y, E(p_x, p_y, \bar{U})) = \bar{U}$$
$$E(p_x, p_y, V(p_x, p_y, I)) = I$$

8.6 Elasticities

Income elasticity:

$$\varepsilon_{x,I} = \frac{\partial x}{\partial I} \cdot \frac{I}{x} = 1$$

(Both goods have unit income elasticity)

Own-price elasticity:

$$\varepsilon_{x,p_x} = \frac{\partial x}{\partial p_x} \cdot \frac{p_x}{x} = -1$$

Cross-price elasticity:

$$\varepsilon_{x,p_y} = \frac{\partial x}{\partial p_y} \cdot \frac{p_y}{x} = 0$$

(Cobb-Douglas has zero cross-price elasticity)

Slutsky equation in elasticity form:

$$\varepsilon_{x,p_x} = \varepsilon^c_{x,p_x} - s_x \varepsilon_{x,I}$$

$$-1 = \varepsilon^c_{x,p_x} - \frac{\alpha}{\alpha + \beta} \cdot 1$$

Therefore:

$$\varepsilon^c_{x,p_x} = -\frac{\beta}{\alpha + \beta}$$

9. Numerical Example

Let's work through a complete example with specific numbers:

- $\alpha = 0.5, \beta = 0.5$ (symmetric preferences)
- $p_x = 10, p_y = 5$
- $I = 100$

9.1 Marshallian Demands

$$x^* = \frac{0.5 \times 100}{(0.5 + 0.5) \times 10} = \frac{50}{10} = 5$$

$$y^* = \frac{0.5 \times 100}{(0.5 + 0.5) \times 5} = \frac{50}{5} = 10$$

Budget check: $10(5) + 5(10) = 50 + 50 = 100$

9.2 Maximum Utility

$$U^* = (5)^{0.5}(10)^{0.5} = \sqrt{50} \approx 7.071$$

9.3 Indirect Utility

$$K = \frac{0.5^{0.5} \times 0.5^{0.5}}{1^1} = 0.5$$

$$V = 0.5 \times \frac{100^1}{10^{0.5} \times 5^{0.5}} = 0.5 \times \frac{100}{\sqrt{50}} = \frac{50}{\sqrt{50}} = \sqrt{50} \approx 7.071$$

9.4 Hicksian Demands (at utility level $\bar{U} = 7.071$)

$$x^h = \left(\frac{0.5}{0.5}\right)^{0.5} \left(\frac{5}{10}\right)^{0.5} (7.071)^1 = 1 \times 0.707 \times 7.071 = 5$$

$$y^h = \left(\frac{0.5}{0.5}\right)^{0.5} \left(\frac{10}{5}\right)^{0.5} (7.071)^1 = 1 \times 1.414 \times 7.071 = 10$$

As expected, Hicksian and Marshallian demands coincide at the optimal income level.

9.5 Expenditure Function

$$E(10, 5, 7.071) = C \times 10^{0.5} \times 5^{0.5} \times (7.071)^1$$

The constant C is chosen so that $E = 100$ at these prices and utility level.

9.6 Slutsky Decomposition

Suppose p_x increases from 10 to 12. What happens to demand for x?

New Marshallian demand:

$$x_{new} = \frac{0.5 \times 100}{1 \times 12} = 4.167$$

Change: $\Delta x = 4.167 - 5 = -0.833$

Substitution effect (hold utility constant at 7.071):

$$x^h(12, 5, 7.071) = 1 \times \left(\frac{5}{12}\right)^{0.5} \times 7.071 = 0.645 \times 7.071 = 4.561$$

$$SE = 4.561 - 5 = -0.439$$

Income effect:

$$IE = 4.167 - 4.561 = -0.394$$

Check: $SE + IE = -0.439 + (-0.394) = -0.833$

Both effects work in the same direction (normal good), so the total effect is the sum.

10. Exercises

Work through these to verify your understanding:

Exercise 1

Derive all the results above for the utility function:

$$U(x, y) = x^{0.3}y^{0.7}$$

What are the expenditure shares now? Are they still constant?

Exercise 2

For the Cobb-Douglas function, verify that:

$$\frac{\partial^2 E}{\partial p_x^2} = \frac{\partial x^h}{\partial p_x} < 0$$

This shows that the expenditure function is concave in prices.

Exercise 3

Show that for Cobb-Douglas, the compensated own-price elasticity is:

$$\varepsilon^c_{x,p_x} = -\frac{\beta}{\alpha + \beta}$$

and verify the elasticity form of the Slutsky equation.

Exercise 4

Calculate the compensating variation when the price of x increases from p_x^0 to p_x^1 for Cobb-Douglas preferences.

$$CV = E(p_x^1, p_y, U_0) - E(p_x^0, p_y, U_0)$$

Compare this to the change in consumer surplus using the Marshallian demand curve.

Exercise 5

For the numerical example in Section 9, calculate:

a) The compensating variation if p_x rises from 10 to 15
b) The equivalent variation for the same price change
c) The consumer surplus loss (area under Marshallian demand)

Compare the three welfare measures.

References

Textbook:

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  - Chapter 4: Utility Maximization and Choice
  - Chapter 5: Income and Substitution Effects

Additional References:

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